DIGITAL VLSI DESIGN

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DIGITAL VLSI DESIGN
Text Book

Book Type	Author & Title	Edition	Publisher	Year
Text book 1	Sung-Mo (Steve) **Kang**, Yusuf Leblebigi, **"CMOS Digital Integrated Circuits Analysis and Design"**	Third	Tata McGraw-Hill Education	2003
Text book 2	Neil Weste and David Harris," CMOS VLSI Design", Pearson Education, 3rd Edition, 2006.	First	Pearson Education	2006

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DIGITAL VLSI DESIGN

PART 1

MOS Inverters (Static Characteristics)

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MOS Inverters (Static Characteristics)

UNIT 1 (Part 1) : MOS Inverters (Static Characteristics)
VLSI design Methodologies and Design Flow (1.4 &1.5)
Introduction to inverters (5.1)
Resistive-Load Inverter (5.2)
Inverters with n-Type MOSFET Load (5.3)
CMOS Inverter (5.4)

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MOS Inverters (Static Characteristics)

Part a - VLSI design Methodologies and Design Flow

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MOS Inverters (Static Characteristics)

Overview of Design Methodology

- **Design Complexity of logic chip** increases almost exponentially with number of transistors to be integrated. This intern increases the design time.

- **Design Cycle Time:** It is the time period from **start of chip development until the mask type delivery.** Most time is used to achieve the desired level of performance at a acceptable cost.

- During the course of design cycle the circuit performance can **be increased by adopting improved design.** This **happens more rapidly in the beginning and then gradually until the performance saturates** for **design style (Full/Semi Custom) and technology used**.

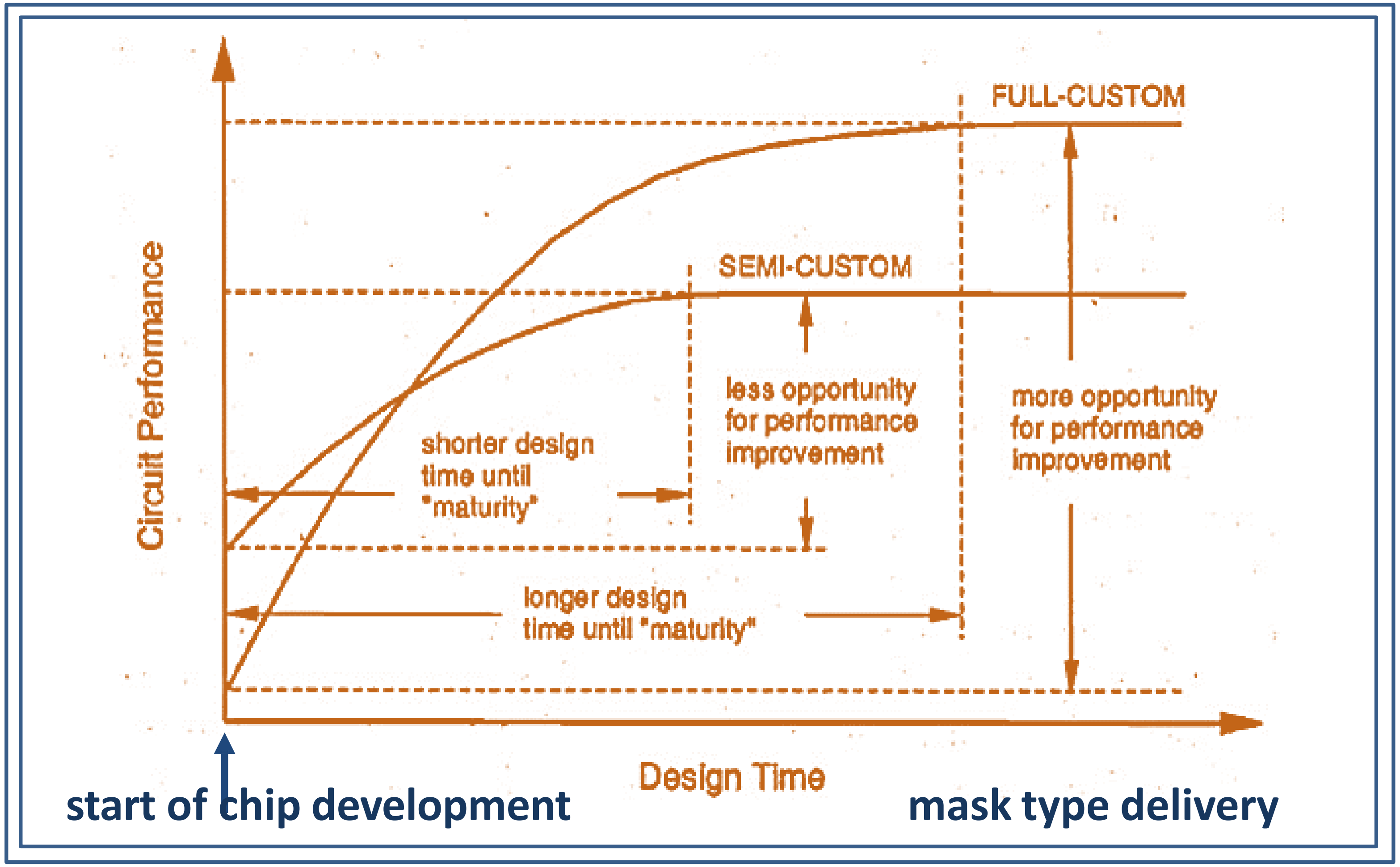
MOS Inverters (Static Characteristics)

Overview of Design Methodology

- **Design Cycle Time and Achievable circuit performance** depends on VLSI Design Style used. The design styles used are

  - ✓ **Full Custom Design style :** Geometry and the placement of every transistor can be optimized individually.
  - ✓ **Semi- Custom Design style** : Use Standard cells / FPGA

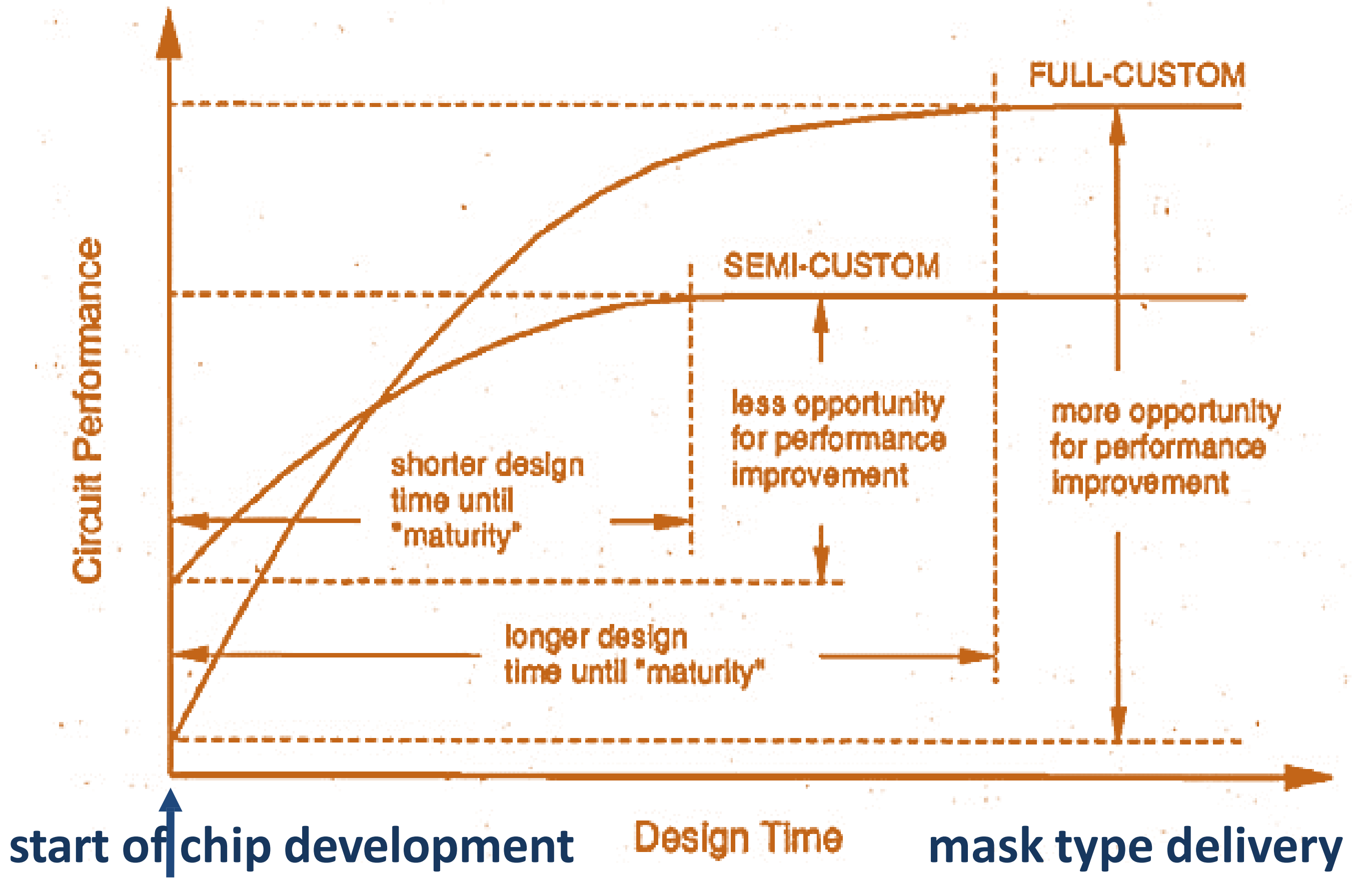
MOS Inverters (Static Characteristics)
Effect of design style on design cycle time and performance



Impact of Different VLSI design Styles on Cycle time

MOS Inverters (Static Characteristics)

Effect of design style on design cycle time and performance



Impact of Different VLSI design Styles on Cycle time

Full Custom :

1.It requires **longer time until the design maturity reached**

2.The flexibility in geometry and placement allows **more opportunity for performance improvement.**

Final product has higher level of performance.

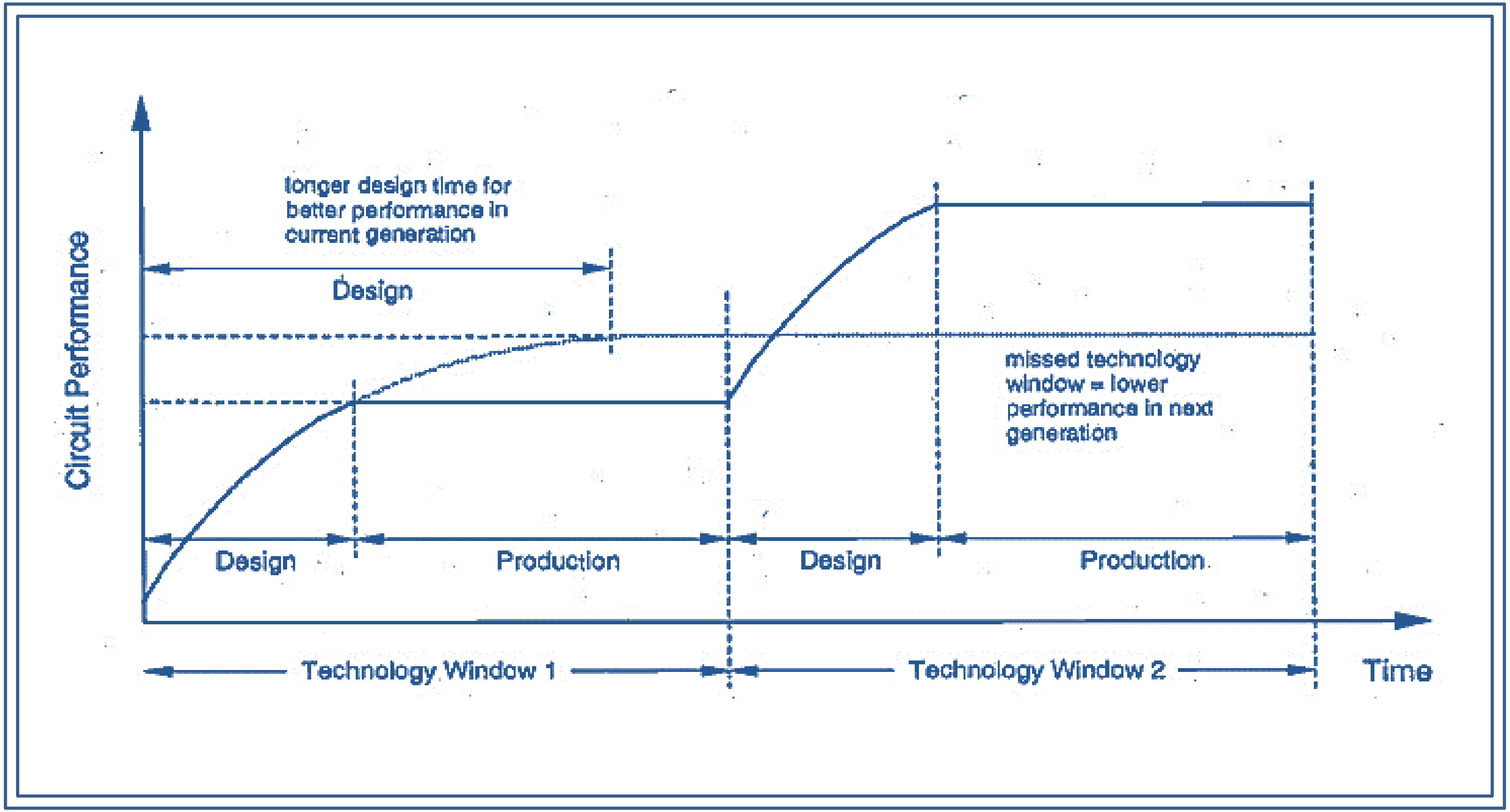
3.Silicon area is relatively small because of better area utilization

Semi Custom:

1. **Less design Time** as it reaches maturity early & **less opportunity for performance improvement**
2. At the **early design phase, the design performance is Higher than Full Custom**

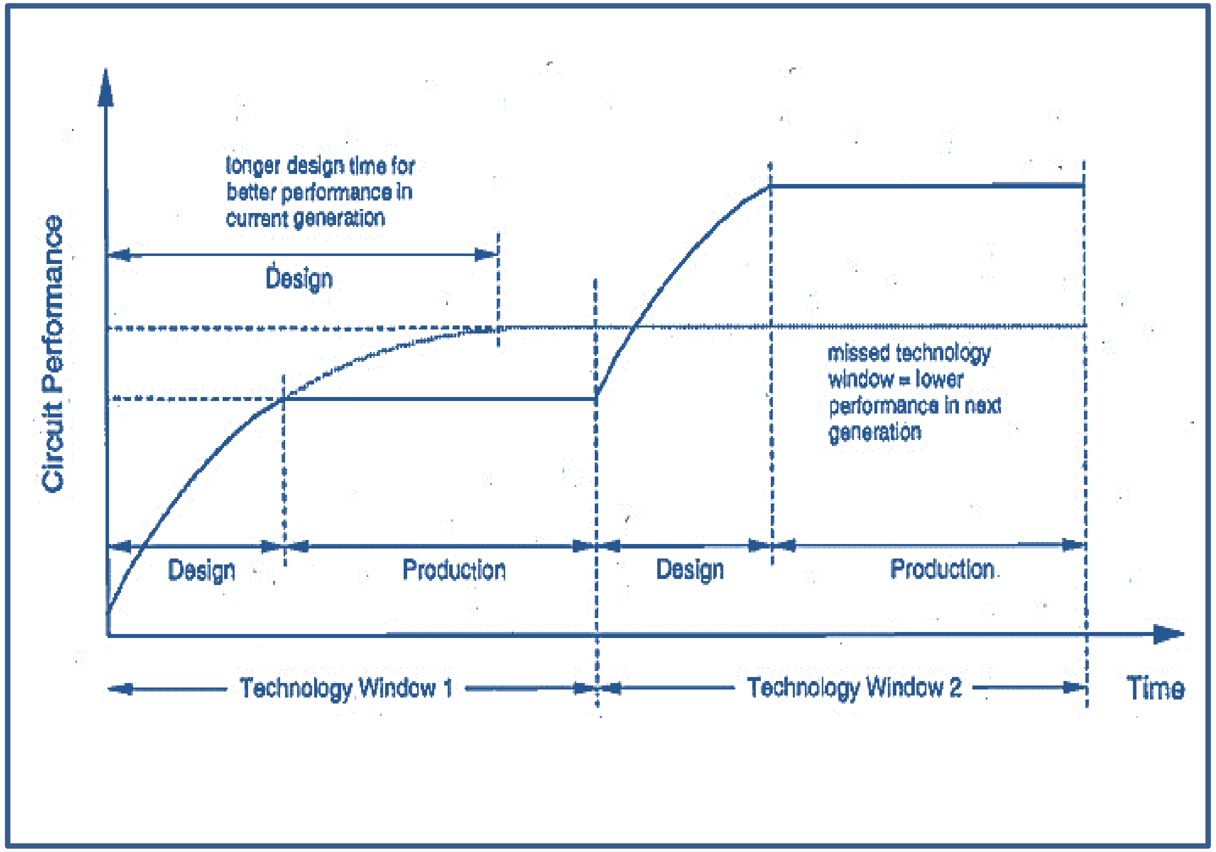
MOS Inverters (Static Characteristics)

Effect of Technology Window on Design Time and Performance



MOS Inverters (Static Characteristics)

Effect of Technology Window on Design Time and Performance



- Every 2 years new generation technology is introduced which typically allows smaller device dimensions, higher integration and performance.
- **To make best use of current technology the chip development time has to be short enough** to allow maturing of chip manufacturing and timely delivery of product to customers

MOS Inverters (Static Characteristics)

VLSI Design Flow : The Y- Chart/Gajski- Chart

A Y-Chart is **a three-part graphic organizer that is used for describing three aspects of a topic**.

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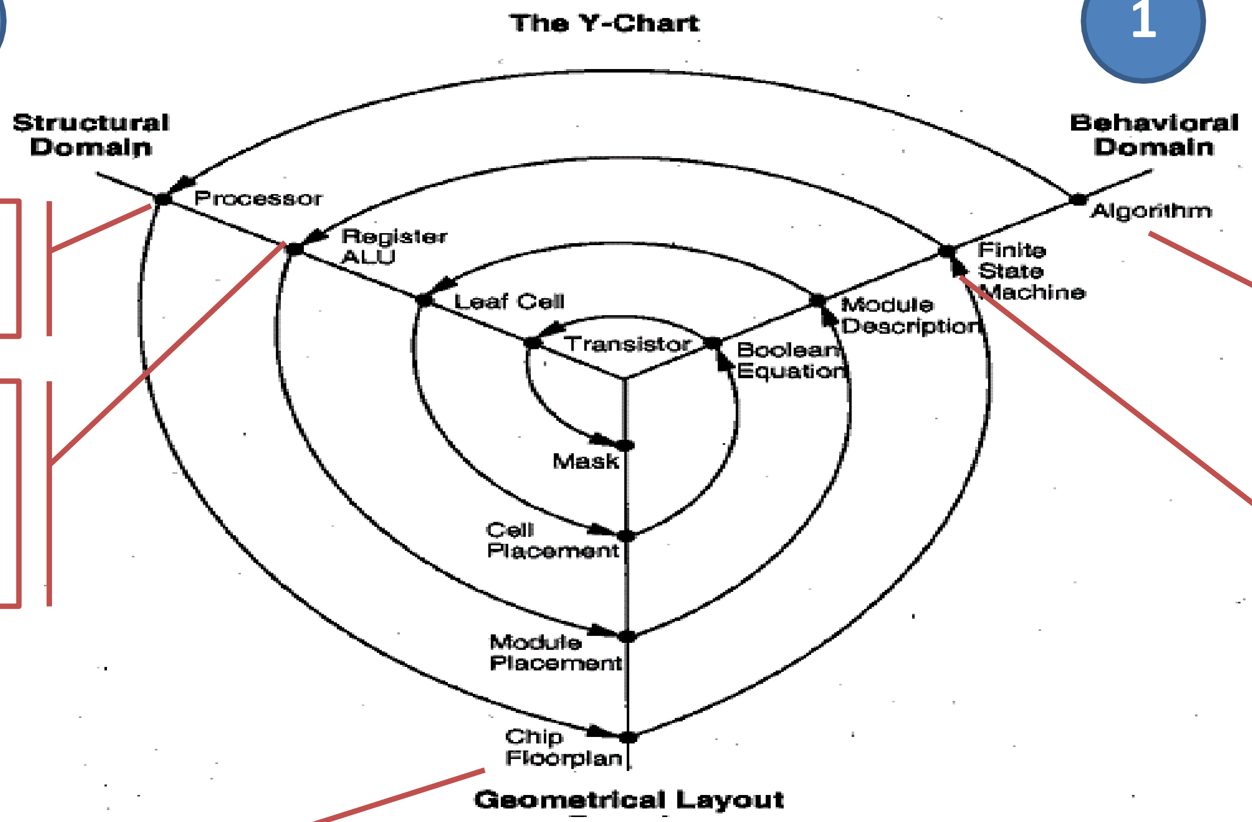


3

Followed by Architecture of Processor

FSM Implemented structurally using FUNCTIONAL Modules as Registers and ALU

Design Flow Starts from the Algorithm that describes the behaviors of target chip

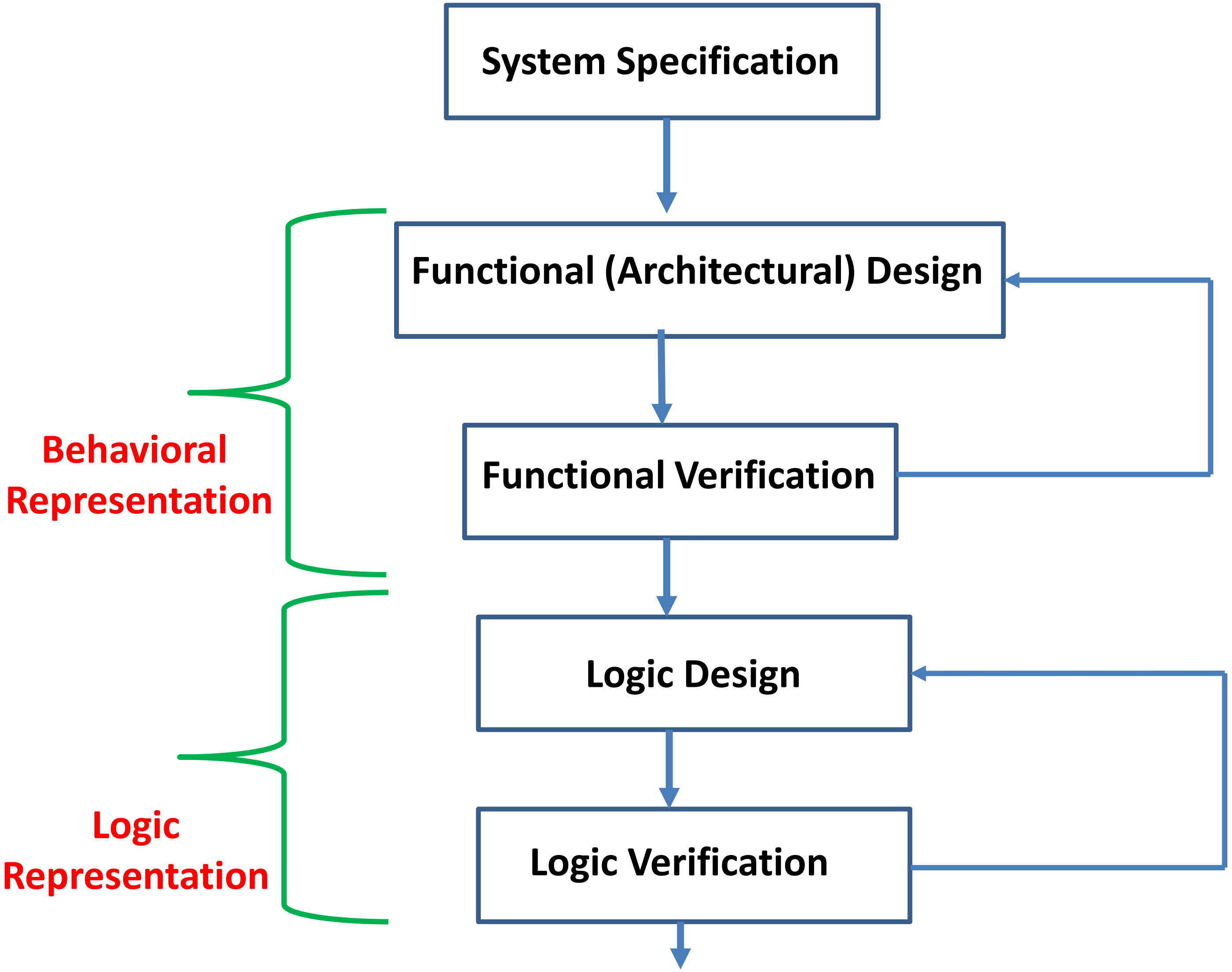
FSM

Mapped onto chip by floor planning

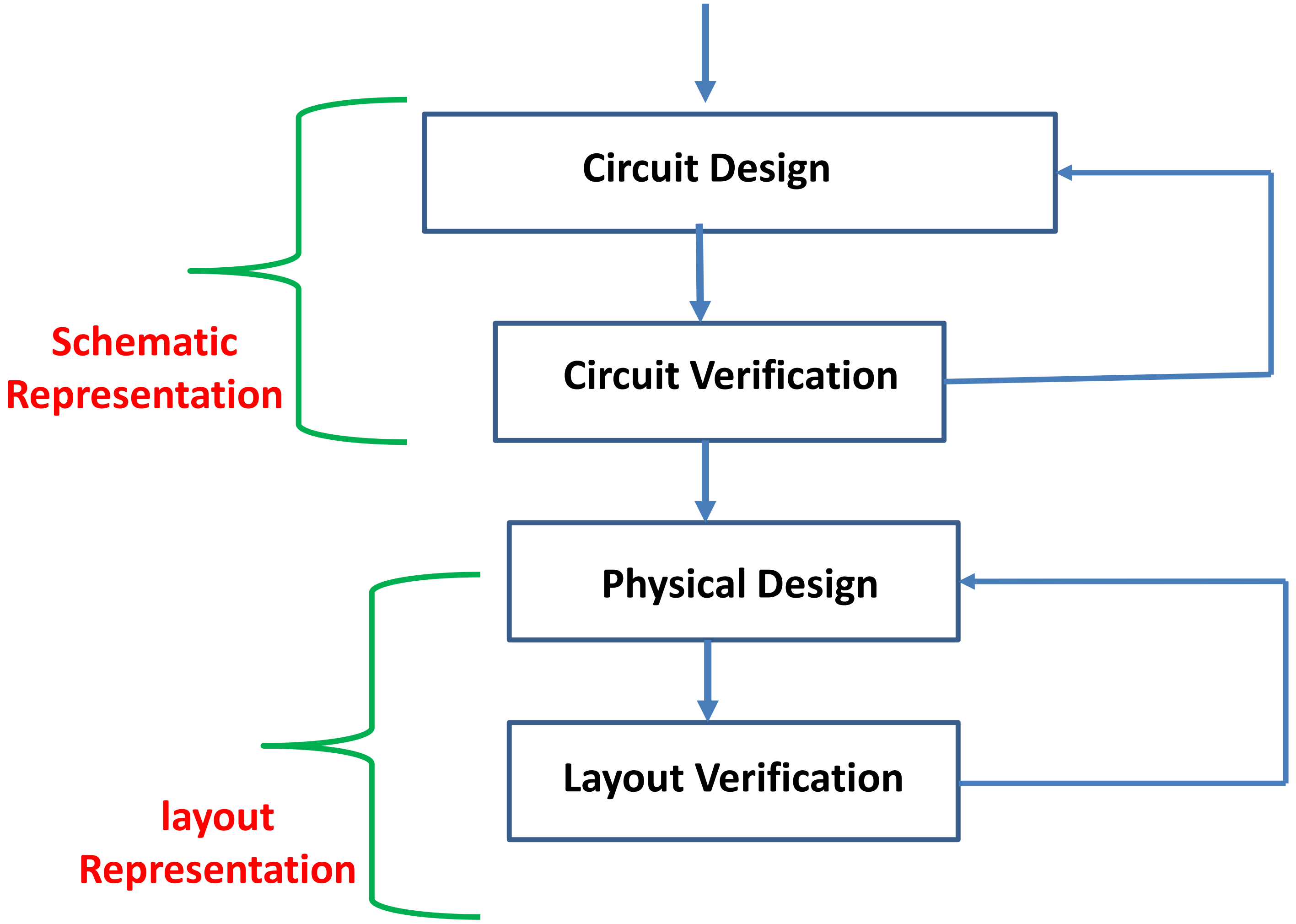
Leaf Cell: Realizing simple logic functionality using small number of Transistors.

MOS Inverters (Static Characteristics)

VLSI Design Flow (ASIC Design Flow)



MOS Inverters (Static Characteristics)
VLSI Design Flow (ASIC Design Flow)



THANK YOU

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